

JEE CHEMISTRY REACTION MAPS

Organic Synthesis Paths & Coordination Compound Split Kinetics

1. Key Organic Reactions & Pathways

Grignard Reagents (RMgX): Organometallic intermediates acting as highly polar nucleophiles. They react with carbonyl groups to build carbon-carbon bonds.

Active Hydrogens (Acid-Base)	$\text{RMgX} + \text{H}_2\text{O} \rightarrow \text{RH} + \text{Mg(OH)X}$
Formaldehyde Addition	$\text{RMgX} + \text{HCHO} \rightarrow \text{Primary Alcohol (1 Degree)}$
Other Aldehydes	$\text{RMgX} + \text{R}'\text{CHO} \rightarrow \text{Secondary Alcohol (2 Degree)}$
Ketones Addition	$\text{RMgX} + \text{R}'\text{COR}'' \rightarrow \text{Tertiary Alcohol (3 Degree)}$
Carbon Dioxide Addition	$\text{RMgX} + \text{CO}_2 \rightarrow \text{R-COOH (Carboxylic Acid)}$

Aldol & Cannizzaro Condensations: Carbonyl coupling pathways dictated by the presence of acidic alpha-hydrogen atoms.

Aldol (with alpha-H)	$2 \text{ R-CH}_2\text{-CHO} + \text{dil. NaOH} \rightarrow \text{beta-hydroxy aldehyde}$
Cannizzaro (no alpha-H)	$2 \text{ HCHO} + \text{conc. NaOH} \rightarrow \text{CH}_3\text{OH} + \text{HCOONa}$ [Disproportionation]

Electrophilic Aromatic Substitutions: Reagents reacting with aromatic benzene rings.

Benzene Nitration	$\text{C}_6\text{H}_6 + \text{HNO}_3/\text{H}_2\text{SO}_4 \rightarrow \text{C}_6\text{H}_5\text{-NO}_2 + \text{H}_2\text{O}$
Friedel-Crafts Alkylation	$\text{C}_6\text{H}_6 + \text{R-Cl} + \text{anhydrous AlCl}_3 \rightarrow \text{C}_6\text{H}_5\text{-R} + \text{HCl}$
Friedel-Crafts Acylation	$\text{C}_6\text{H}_6 + \text{R-CO-Cl} + \text{anhydrous AlCl}_3 \rightarrow \text{C}_6\text{H}_5\text{-CO-R} + \text{HCl}$

2. Nucleophilic Substitutions (SN1 vs SN2)

SN1 Rate Kinetics	$\text{Rate} = k * [\text{Substrate}]$ [2-Step, Carbocation Intermediate]
SN2 Rate Kinetics	$\text{Rate} = k * [\text{Substrate}] * [\text{Nu-}]$ [1-Step, Walden Inversion]

SN1 pathways are favored by polar protic solvents, tertiary structures, and stable carbocations. SN2 pathways are favored by polar aprotic solvents, primary structures, and low steric hindrance.

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Part 2: Crystal Field Split Energies & Kinetic Stability

3. Crystal Field Theory (CFT)

Crystal Field Splitting (d-Orbitals): The electrostatic approach of ligands lifts the degeneracy of metal d-orbitals. In octahedral fields, the energy splits into lower t_{2g} and upper e_g levels.

Octahedral Splitting (Do)	Splits into t _{2g} (d _{xy} , d _{yz} , d _{xz}) & e _g (d _{x²-y²} , d _{z²})
Tetrahedral Splitting (Dt)	Dt = (4/9) * Do [e is lower energy, t ₂ is higher]
CFSE calculation (Oct)	CFSE = [-0.4 * n(t _{2g}) + 0.6 * n(e _g)] * Do + m * P

Where n(t_{2g}) and n(e_g) are electron populations, Do is the splitting constant, P is the pairing energy, and m is the number of electron pairs in the d-orbitals.

4. Kinetic & Thermodynamic Stability

Stepwise Formation Const.	K _n = [ML _n] / ([ML _(n-1)] * [L])
Overall Stability (Beta _n)	Beta _n = K ₁ * K ₂ * ... * K _n

A high thermodynamic stability constant (log Beta) denotes a strong ligand-metal coordinate interaction. High kinetic inertness signifies slow rate exchange of ligands (e.g., d³ or low-spin d⁶ complexes).